Fiscal Year:	FY 2026	**Task Last Updated:**	FY 11/24/2025
PI Name:	Macaulay, Timothy Ph.D.		
Project Title:	Neuro-Vestibular Examination During and After Spaceflight (Vestibular Health) (PI: Macaulay)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	KBR/NASA Johnson Space Center		
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City:	Houston	**State:**	TX
Zip Code:	77058-3711	**Congressional District:**	36
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C
Start Date:	01/30/2019	**End Date:**	09/30/2033
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Project start date entered as 01/30/2019 to match original investigation, as requested by NASA HHC (Ed., 1/3/25).		
Key Personnel Changes/Previous PI:	Per the Principal Investigator, Drs. Schubert, Shelhamer, and Kheradmand were added to the project team (Ed., 1/3/26).		
COI Name (Institution):	Dervay, Joseph M.D. (NASA Johnson Space Center) Makishima, Tomoko M.D., Ph.D. (University of Texas Medical Branch at Galveston) Wood, Scott Ph.D. (NASA Johnson Space Center) Clement, Gilles Ph.D. (KBR/NASA Johnson Space Center - Retired) Schubert, Michael (Johns Hopkins University) Shelhamer, Mark (Johns Hopkins University) Kheradmand, Amir (Johns Hopkins University School of Medicine)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	Adaptation to the absence of Earth's gravitational environment during spaceflight causes neurological disturbances that are either directly or indirectly mediated by the vestibular system. These disturbances include space motion sickness, spatial disorientation, and cognitive impairment, as well as changes in head-eye coordination, vestibulo-ocular reflex, and interactions with support surfaces. After return to Earth, astronauts experience other vestibular-driven behavioral changes, including re-entry motion sickness, motion illusions, gaze-induced nystagmus, and balance and locomotion deficits. Otolith-mediated reflex gain changes are striking shortly after g-transitions. However, animal studies have shown that structural modifications of the vestibular sensory apparatus may occur throughout an extended spaceflight exposure. To date, no long-duration flight studies have directly investigated potential changes in the vestibular organs of astronauts. As a part of the Complement of Integrated Protocols for Human Research (CIPHER), this study will address this gap by performing a systematic neuro-vestibular examination of crewmembers in orbit at regular intervals, as well as immediately after landing. Crewmembers will be tested at standardized timepoints to investigate temporal changes, and to identify trends in adaptation in vestibular health and performance. In orbit, subjects will perform active head movements and be exposed to various passive maneuvers executed by the operator. Observations and recordings of eye, head, and body movements, as well as subjective perception of motion and verbal reports, will be used for evaluating the presence of abnormal eye movements, dysmetria, motion sickness symptoms, and illusions of motion during head or body movements. Tests will be performed both early in the mission and once every one or two months thereafter. The post-flight examination will be performed on the day of landing (R+0) and several times thereafter. During one preflight session, a simple and rapid measure of otolith asymmetry – a basic low-level neurovestibular function that may be altered in spaceflight – will also be administered. This provides a measure of binocular alignment (vertical and torsional), which reflects underlying otolith asymmetry. This testing provides an opportunity to assess a low-level neural system that is not under volitional control, is responsive to g-level alterations, and might be an indicator of broader and more functionally relevant effects including inflight/postflight measures of (mal)adaptation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The tests proposed in this study are well established and validated protocols that can detect acute or chronic vestibular syndromes. Repeated measurements during adaptation to g-transitions will provide insight into the vestibular adaptation processes. If the observed physiological changes in crewmembers are more deleterious after longer duration missions on the International Space Station (ISS), then relevant countermeasures will be required to enable longer duration missions. Depending on the etiology of the vestibular disorders revealed by our tests, monitoring for long-term health outcomes and vestibular rehabilitation countermeasures can be tailored to the deficits observed.
Task Progress:	This past reporting period, the study team continued collecting data on International Space Station (ISS) crewmembers as part of the Complement of Integrated Protocols for Human Exploration Research (CIPHER) study. These included 3 preflight, 9 inflight, and 4 postflight data collections. These were accomplished despite the challenges of spaceflight operations. For example, two in-flight sessions were repeated due to data dropout issues. An in-flight functional checkout was completed to identify the causes and mitigations for these issues. To date, four crewmembers have completed all preflight, in-flight, and post-flight testing for this study. Automated data processing routines for eye tracking data have been developed and consolidated into a single MATLAB application with a unified Graphical User Interface (GUI). This development allows comparison of different eye tracking processing algorithms in a modular approach that can be shared across the different study-specific aims. The modular approach increases the efficiency with which new analysis tools can be implemented by building on the same calibration and GUI framework. Eye tracking data have been processed for several key measures. The Vestibular Ocular Reflex (VOR) tests measure gaze stabilization during active head movements. VOR gain was calculated both in the dark and while subjects fixated on a visual suppression target. Various nystagmus tests were performed to quantify drifting eye movements and nystagmus while subjects were in the dark and instructed to look either straight ahead or at imaginary 15-degree targets. Saccadic eye function was assessed from subjects capturing 5-degree, 10-degree, and 15-degree visual wall targets while keeping their heads still. Body movement perception data have also been processed, including perceptions of whole-body passive rotations and translations in microgravity and blindfolded walking/stepping measures on Earth for the perceptions of egocentric distance, egocentric rotation, and path integration. Plots of these preliminary processed data have been generated based on their timeline of completion relative to launch and landing. Testing sessions include 90 days prior to launch (L-90), flight day 1 (FD1), flight day 30 (FD30), 30 flight days before return (FDR-30), return day (R+0), 4 days after return (R+4), 9 days after return (R+9), and 30 days after return (R+30).
Bibliography Type:	Description: (Last Updated: 12/23/2025)
Articles in Peer-reviewed Journals	Kuldavletova O, Navarro Morales DC, Macaulay TR, Wood SJ, Toupet M, Hautefort C, Van Nechel C, Rengel A, Kola A, Quarck G, Denise P, Clément G. "Bilateral vestibulopathy affects spatial and temporal perception." PLoS One. 2025 Nov 6;20(11):e0336108. https://doi.org/10.1371/journal.pone.0336108 ; PubMed PMID: 41196860; PubMed Central PMCID: PMC12591442 , Nov-2025